Linear Functional (Row Vector)

$\lambda : \mathbb{R}^n \to \mathbb{R}^1$ linear transformation.

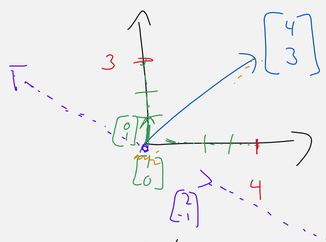
$1 \times n$ matrix $\quad [a_1 \cdots a_n]$

$$\lambda(v) = [a_1 \cdots a_n] \begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix} = [w \cdot v]$$

$\lambda = w'$, v

A linear functional creates a "coordinate map"

A basis of linear functionals creates a coordinate system.



x-coord $\left(\begin{bmatrix} 4 \\ 3 \end{bmatrix}\right) = 4$ $\quad [1 \ 0] \begin{bmatrix} a \\ b \end{bmatrix} = a$

y-coord $\left(\begin{bmatrix} 4 \\ 3 \end{bmatrix}\right) = 3$ $\quad [0 \ 1] \begin{bmatrix} a \\ b \end{bmatrix} = b$

x-y coord. system

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} a \\ b \end{bmatrix}$$

$2(4) - 1(3) = 8 - 3 = 5$

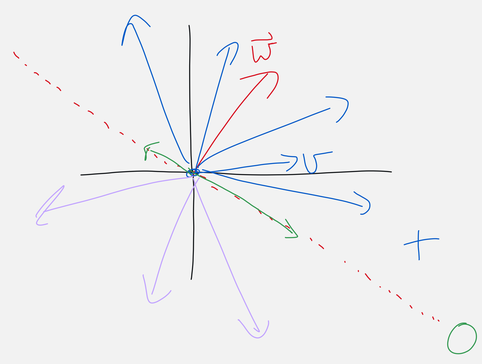
$$[2 \ -1] \begin{bmatrix} a \\ b \end{bmatrix} = 2a - b$$

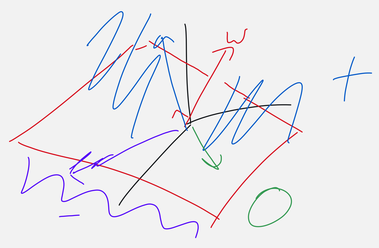
+ and − values of functionals:

$w'(v) > 0$ $\quad w, v$ acute

$w'(v) = 0$ $\quad w, v$ orthogonal

$w'(v) < 0$ $\quad w, v$ obtuse angle.



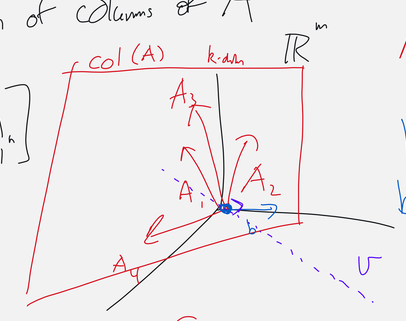


(2.1)

A: $\quad im(A) = col(A)$

span of columns of A

$$A = \begin{bmatrix} | & & | \\ A_1 & \cdots & A_n \\ | & & | \end{bmatrix}$$



$Ax = b$

$b \in col(A)$ → solvable

$b \notin col(A)$ → no solution

$$A\vec{x} = \begin{bmatrix} | & & | \\ A_1 & \cdots & A_n \\ | & & | \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$$

$col(A) = \{A\vec{x} : \vec{x} \in \mathbb{R}^n\}$

$= x_1 A_1 + x_2 A_2 + \cdots + x_n A_n$

If $b \in col(A)$, then $Ax = b$ has solution.

If $b \notin col(A)$, then we want to find $\vec{v}$

$$\text{s.t.} \begin{cases} \vec{v}' A = 0 \\ \vec{v}' b \neq 0 \end{cases}$$

$$\vec{v}' A = \begin{bmatrix} — & v' & — \end{bmatrix} \begin{bmatrix} | & & | \\ A_1 & \cdots & A_n \\ | & & | \end{bmatrix} = [v \cdot A_1 \ \ v \cdot A_2 \ \cdots \ v \cdot A_n] = [0 \ 0 \ \cdots \ 0]$$

if perpendicular

So, Necessary that $\vec{v} \in (col A)^\perp$

so, if $col(A) = \mathbb{R}^m$, $(col A)^\perp = 0$, and also $b \in col(A)$ ✓ (solvable)

else, $b \in col(A)$, then $col(A) \neq \mathbb{R}^m$, $(col A)^\perp \neq 0$, so $v \in (col A)^\perp$ exists $(\neq 0)$

BARUCH COLLEGE, CUNY

SOME NOTES ON LP AND OPTIMAL TRANSPORT

GIULIO TRIGILA

Abstract. This is a set of informal notes introducing to linear programming (LP) and showing how this is connected to the discrete Kantorovich problem in optimal transport. There is nothing new here, the LP part is taken from the Linear Algebra book by Peter Lax. Please let me know if you think something is not making sense and feel free to suggest improvements to this document.

1. Linear Programming

In this section we will review the fundamentals of the theory of duality in linear programming. We will start showing how the theorem of the alternative for linear systems is generalized when inequality constraints are added. This will be useful to prove the duality principle. We will proof only the main theorems referring to the literature when a proof is not provided.

Every vector is intended to be a column vector, while row vectors are marked with an apex. Apices are also used to indicate the transpose of a matrix.

2. Theorem of the alternative

Let $A \in \mathbb{R}^{m \times n}$, $x \in \mathbb{R}^n$ and $b \in \mathbb{R}^m$. The solvability of the linear system $Ax = b$ is characterized by the following theorem

Theorem 2.1. *(Fredholm). Either the linear system $Ax = b$ has a solution or exists v such that $v'A = 0$ and $v'b \neq 0$.*

In words either b is in the column space of A or otherwise b has a nonzero component in the null space of A'.

The theorem (2.1) can be generalized in the case we are requiring the solution of $Ax = b$ to be (component-wise) non negative

Theorem 2.2. *(Farkas). Either the problem*

$$\begin{cases} Ax = b \\ x \geq 0 \end{cases} \tag{2.3}$$

has a solution or there exists a v such that $v'A \geq 0$ and $v'b < 0$.

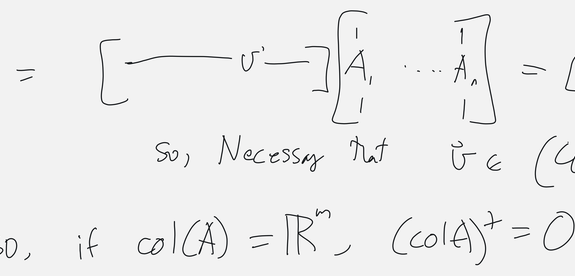
Date: July 3, 2022.

(2.2)

$$\begin{cases} Ax = b \\ x \geq 0 \end{cases}$$

$$Ax = x_1 \begin{bmatrix} | \\ A_1 \\ | \end{bmatrix} + \cdots + x_n \begin{bmatrix} | \\ A_n \\ | \end{bmatrix}$$

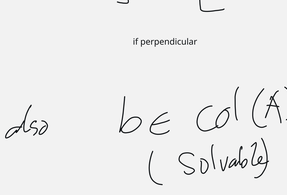
L.C. of cols of A, but only nonnegative x_i



(2.4) Cone is Convex and Closed

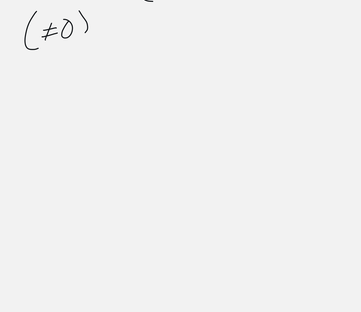
↳ ∴ Given a point y not in the set S, then is a closed half-space containing S and not y.

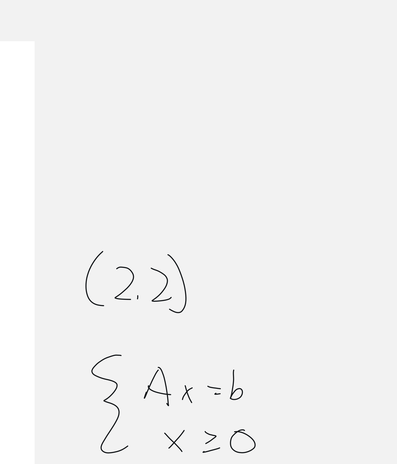
Convex Function



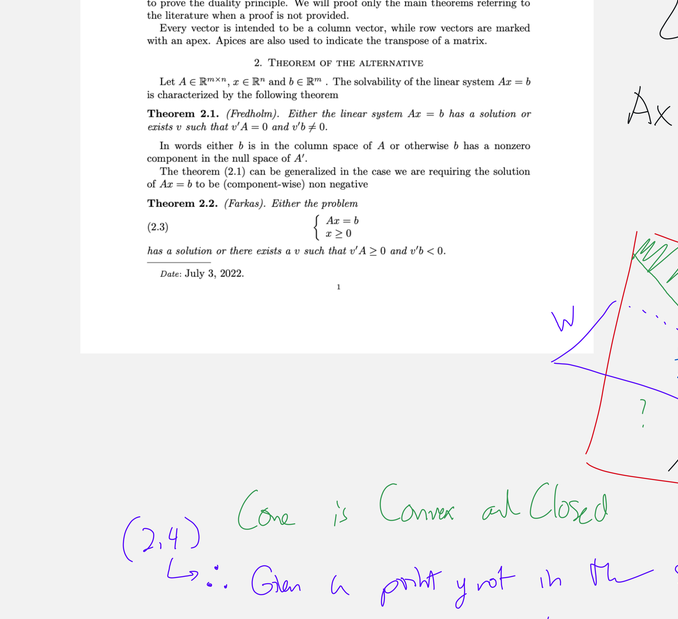
Convex Space

A convex function defines a convex space above the graph





2.4



Because of theorem (2.4) either b is in K, namely system (2.3) has a solution, or otherwise it is possible to find an hyperplane that goes through the origin and that has the vector b on one side and the set K on the other. Let v be the vector perpendicular to such plane. Then the angle in between v and any vector in K will be less or equal to $\pi/2$, namely $v'A \geq 0$ while the angle in between v and b is going to be bigger than $\pi/2$ resulting in $v'b < 0$.

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

$$v'A = \begin{bmatrix} v \cdot A_1 & v \cdot A_2 \end{bmatrix} \geq 0$$

$$v'b = \begin{bmatrix} v \cdot b \end{bmatrix} < 0$$

2.4 gives the intuition to understand 2.2

closed half space is purple
think of it as a plane with orientation